Graphic Era HILL UNIVERSITY
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University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH, ME V SEMESTER
Name of the Paper: HEAT TRANSFER
Paper Code: TME 501
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Explain Stefan-Boltzmann Law of radiation heat transfer. How does emissivity affect radiation heat transfer?

OR

b. What is combined heat transfer? Explain with an example how conduction, convection, and radiation occur together in a system.

Q2. (10 Marks)

a. A wall is 25 cm thick with a thermal conductivity of 0.8 W/mK. The temperature on one side of the wall is 100°C and on the other side is 25°C. The surface area of the wall is 10 m^2.

Calculate:

i. The rate of heat transfer through the wall.
ii. The temperature gradient within the wall.

OR

b. A composite cylindrical wall consists of two materials. Material 1 has an inner radius of 0.05 m, outer radius of 0.07 m, and thermal conductivity k_1=0.5 W/mK. Material 2 has an inner radius of 0.07 m, outer radius of 0.1 m, and thermal conductivity k_2=0.2 W/mK. The inside temperature is 150°C and the outside temperature is 50°C.

Calculate:

i. The rate of heat transfer per meter length of the cylindrical wall.

Q3. (10 Marks)

a. Derive the general three-dimensional heat conduction equation in Cartesian coordinates.

OR

b. Explain the heat transfer characteristics of a short fin with an insulated tip. Derive the equation for heat transfer and discuss boundary conditions.

Q4. (10 Marks)

a. A steel fin with a uniform cross-sectional area of 0.5 cm^2 is 6 cm long and has an insulated tip. The base temperature is 200°C, and the ambient temperature is 25°C. The heat transfer coefficient is h=35 W/m^2K, and the thermal conductivity of steel is k=45 W/mK.

Calculate:

i. The fin efficiency.
ii. The rate of heat transfer from the fin.

OR

b. A fin made of aluminum (thermal conductivity k=205 W/mK is connected between two heat sources. The temperature of the heat sources at either end of the fin is 200°C, respectively. The fin is 10 cm long with a cross-sectional area of 2 cm^2. The ambient temperature is 25°C, and the heat transfer coefficient is 30 W/m^2.

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Calculate:

i. The temperature distribution along the fin.
ii. The heat transfer rate from the fin.

Q5. (10 Marks)

a. Discuss the heat transfer and temperature distribution in a short fin with a non-insulated (convective) tip. Present the corresponding heat transfer equation and boundary conditions.

OR

b. How does the length of a fin affect the rate of heat transfer? Compare the performance of a long fin versus a short fin.

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